

2. Perform basic Image Handling and processing operations on the image Read an image in Python and Convert an Image to Blur using Gaussian Blur.

Aim:

To perform basic image handling operations by reading an image in Python and applying Gaussian Blur to reduce noise and smooth the image using the OpenCV library.

Procedure:

- Import the required Python libraries (OpenCV and Matplotlib).
- Read the input image using the cv2.imread() function.
- Apply Gaussian Blur to the image using the cv2.GaussianBlur() function with a 5×5 kernel.
- Display the original image and the Gaussian blurred image.
- Observe the smoothing effect and reduction of noise in the blurred image.

Code:

```
import cv2

import matplotlib.pyplot as plt

# Read the image

image = cv2.imread("/Users/michellejoanna/Downloads/meee.jpeg")

# Convert the image from BGR to RGB

rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Apply Gaussian Blur

blur = cv2.GaussianBlur(rgb, (15, 15), 0)

# Display the original and blurred images

plt.figure(figsize=(10, 5))
```

```
plt.subplot(1, 2, 1)
```

```
plt.imshow(rgb)
```

```
plt.title("Original Image")
```

```
plt.axis("off")
```

```
plt.subplot(1, 2, 2)
```

```
plt.imshow(blur)
```

```
plt.title("Gaussian Blurred Image")
```

```
plt.axis("off")
```

```
plt.show()
```

Sample Input:



Sample Output:

Gaussian Blur



Result:

The input image was successfully read using Python and processed using the Gaussian Blur filter.